Mini-Project (ML for Time Series) - MVA 2025/2026

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What is expected for these mini-projects? The goal of the exercise is to read (and understand) a research article, implement it (or find an implementation), test it on real data and comment on the results obtained. Depending on the articles, the task will not always be the same: some articles are more theoretical or complex, others are in the direct line of the course, etc... It is therefore important to balance the exercise according to the article. For example, if you have reused an existing implementation, it is obvious that you will have to develop in a more detailed way the analysis of the results, the influence of the parameters etc... Do not hesitate to contact us by email if you wish to be guided.

The report The report must be at most FIVE pages and use this template (excluding references). If needed, additional images and tables can be put in Appendix, but must be discussed in the main document. The report must contain a precise description of the work done, a description of the method, and the results of your tests. Please do not include source code! The report must clearly show the elements that you have done yourself and those that you have reused only, as well as the distribution of tasks within the team (see detailed plan below.)

The source code In addition to this report, you will have to send us a Python notebook allowing to launch the code and to test it on data. For the data, you can find it on standard sites like Kaggle, or the site https://timeseriesclassification.com/ which contains a lot of signals!

The oral presentations They will last 10 minutes followed by 5 minutes of questions. The plan of the defense is the same as the one of the report: presentation of the work done, description of the method and analysis of the results.

Deadlines Two sessions will be available :

- **Session 1**
  - Deadline for report: December 14th (23:59)
  - Oral presentations: December 15th and 17th (precise times TBA)
- **Session 2**
  - Deadline for report: January 4th (23:59)
  - Oral presentations: January, 5th and 7th (precise times TBA)

1 Introduction and contributions

Multivariate time series analysis plays a central role in many fields, including econometrics, finance and neuroscience. A widely used modeling framework is the Vector Autoregressive (VAR) model [2], valued for its interpretability and analytical simplicity. However, VAR models rely on linear dynamics with additive, typically Gaussian, innovations, which limits their ability to capture nonlinear interactions, non-stationary volatility and heavy-tailed behavior commonly observed in real-world data, particularly in financial markets. Moreover, Gaussian assumptions hinder the identifiability of latent innovations, complicating causal interpretation.

To overcome these limitations, Morioka et al. [1] introduce the Independent Innovation Analysis (IIA) framework. IIA extends the VAR model to a nonlinear setting in which innovations may interact nonlinearly with past observations. Identifiability is achieved by assuming that latent innovations are mutually independent while being statistically dependent on an auxiliary variable $\mathbf{u}_t$ capturing non-stationarity (e.g., time or segment labels), following principles from nonlinear ICA. The authors propose several estimation strategies, notably IIA-GCL for continuous modulation and IIA-TCL for piecewise-stationary settings.

In this project, we assess the ability of the IIA framework to recover meaningful latent drivers from real financial data. We apply the method to log-returns of major U.S. equities between 2018 and 2023, a setting characterized by regime shifts and volatility clustering that challenge linear models. After an initial dimensionality reduction step using PCA, we compare IIA-based representations to standard linear baselines to evaluate whether modeling nonlinear interactions leads to more interpretable and disentangled market signals.

Contributions and work distribution. The work was distributed as follows: Swann Cordier focused on the implementation and analysis of the **IIA-TCL** algorithm (segment-based), while Baran Celik focused on the **IIA-GCL** algorithm (time-index-based). Regarding the source code, we reused approximately 50% of the original authors' implementation. However, on the simulated data, we went beyond simple reproduction by modifying the code to investigate the influence of key parameters (sensitivity analysis) and assess the method's limitations. Furthermore, we developed a custom data processing pipeline and training scripts to apply these algorithms to a novel task: unsupervised feature extraction from high-dimensional stock market data to analyze non-stationary market dynamics.

2 Method

2.1 Problem setting: nonlinear VAR and innovation recovery

We observe a multivariate time series $\mathbf{x}_t \in \mathbb{R}^n$ generated by a first-order nonlinear vector autoregressive (NVAR) model:

$$\mathbf{x}_t = \mathbf{f}(\mathbf{x}_{t-1}, \mathbf{s}_t) \tag{1}$$

where $\mathbf{f} : \mathbb{R}^{2n} \to \mathbb{R}^n$ is an unknown nonlinear mixing function and $\mathbf{s}_t \in \mathbb{R}^n$ denotes the latent *innovations* (or errors) at time t. Unlike standard VAR models, the interaction between the past observations $\mathbf{x}_{t-1}$ and the new information $\mathbf{s}_t$ can be fully nonlinear, implying that innovations are not necessarily additive. The goal of Independent Innovation Analysis (IIA) is to estimate $\mathbf{s}_t$ from observations only.

To ensure the problem is tractable, the framework considers an augmented formulation. We define

the augmented mapping $\tilde{\mathbf{f}} : \mathbb{R}^{2n} \to \mathbb{R}^{2n}$ as:

$$\begin{bmatrix} \mathbf{x}_t \\ \mathbf{x}_{t-1} \end{bmatrix} = \tilde{\mathbf{f}} \left(\begin{bmatrix} \mathbf{s}_t \\ \mathbf{x}_{t-1} \end{bmatrix} \right) = \begin{bmatrix} \mathbf{f}(\mathbf{x}_{t-1}, \mathbf{s}_t) \\ \mathbf{x}_{t-1} \end{bmatrix} \tag{2}$$

Assuming $\tilde{\mathbf{f}}$ is invertible and smooth, we seek to learn the inverse (demixing) model $\tilde{\mathbf{g}} = \tilde{\mathbf{f}}^{-1}$:

$$\begin{bmatrix} \mathbf{s}_t \\ \mathbf{x}_{t-1} \end{bmatrix} = \tilde{\mathbf{g}} \left(\begin{bmatrix} \mathbf{x}_t \\ \mathbf{x}_{t-1} \end{bmatrix} \right) = \begin{bmatrix} \mathbf{g}(\mathbf{x}_t, \mathbf{x}_{t-1}) \\ \mathbf{x}_{t-1} \end{bmatrix} \tag{3}$$

Estimating the innovations is therefore equivalent to learning the function $\mathbf{g}$, which is parameterized in practice by a neural network $\mathbf{h}_\theta(\mathbf{x}_t, \mathbf{x}_{t-1})$. The theory guarantees identifiability of $\mathbf{s}_t$ up to permutation and component-wise invertible nonlinearities.

2.2 Innovation model with an auxiliary variable

To resolve the indeterminacy inherent in nonlinear mixing, IIA relies on the assumption that the innovations are mutually independent but statistically dependent on an observable auxiliary variable u_t (e.g., time index or segment label). Specifically, conditionally on u_t, the components of $\mathbf{s}_t$ are independent and follow a univariate exponential family distribution:

$$p(\mathbf{s}_t \mid u_t) = \prod_{i=1}^{n} \frac{Q_i(s_{i,t})}{Z_i(u_t)} \exp \left(\sum_{j=1}^{k} q_{ij}(s_{i,t}) \lambda_{ij}(u_t) \right) \tag{4}$$

where q_{ij} are the sufficient statistics, $\lambda_{ij}(u_t)$ are the modulation parameters depending on u_t, and Z_i is the normalizing constant. For identifiability, the modulation matrix $\mathbf{L}$ (constructed from differences in $\lambda(u)$) must be invertible, a condition known as the *Assumption of Variability*.

2.3 IIA-GCL: General Contrastive Learning

The IIA-GCL framework [5] addresses the case where u_t is a general, possibly continuous variable (e.g., the time index t itself). The estimation is formulated as a self-supervised binary classification task. We construct two datasets:

- **Positive samples:** Real tuples $\tilde{\mathbf{x}}_t = (\mathbf{x}_t, \mathbf{x}_{t-1}, u_t)$ drawn from the joint distribution.
- **Negative samples:** Randomized tuples $\tilde{\mathbf{x}}_t^* = (\mathbf{x}_t, \mathbf{x}_{t-1}, u^*)$ where u^* is drawn from the marginal $p(u)$, effectively breaking the dependency between the observations and the auxiliary variable.

We train a nonlinear logistic regression classifier using a regression function $r(\tilde{\mathbf{x}}_t)$ to discriminate between these classes. The function is structured to match the difference of log-pdfs between the joint and product distributions:

$$r(\tilde{\mathbf{x}}_t) = \sum_{i=1}^{n} \sum_{j=1}^{k} \psi_{ij}(h_i(\mathbf{x}_t, \mathbf{x}_{t-1})) \mu_{ij}(u_t) + \phi(\mathbf{x}_{t-1}, u_t) + \alpha(u_t) + \beta(\mathbf{h}(\mathbf{x}_t, \mathbf{x}_{t-1})) + \gamma(\mathbf{x}_{t-1}) \tag{5}$$

where ψ_{ij}, h_i, and μ_{ij} are neural networks approximating sufficient statistics and parameters, while α, β, γ and ϕ account for normalizing constants and marginal densities. Theoretically, as the classifier approaches the optimal Bayes classifier, the learned feature extractor $\mathbf{h}(\mathbf{x}_t, \mathbf{x}_{t-1})$ converges to the true innovations.

2.4 IIA-TCL: Time Contrastive Learning

The IIA-TCL method [4] handles the specific case where u_t takes values in a finite set of classes $\{1, \ldots, T\}$, typically representing time segments (Piecewise Stationarity). The learning problem is framed as a multiclass classification task using Multinomial Logistic Regression (MLR).

Given the observations $(\mathbf{x}_t, \mathbf{x}_{t-1})$, the model learns to predict the correct segment label u_t. The posterior probability is modeled via a softmax function:

$$p(u_t = \tau \mid \mathbf{x}_t, \mathbf{x}_{t-1}) = \frac{\exp\left(\sum_{i=1}^{n} \sum_{j=1}^{k} z_{ij\tau}\right)}{\sum_{l=1}^{T} \exp\left(\sum_{i=1}^{n} \sum_{j=1}^{k} z_{ijl}\right)} \tag{6}$$

where the logits are defined as $z_{ijl} = w_{ijl}\psi_{ij}(h_i(\mathbf{x}_t, \mathbf{x}_{t-1})) + \phi(\mathbf{x}_{t-1}, u_t = l) + b_l$. By maximizing the likelihood of the true labels, the feature extractor $\mathbf{h}$ is forced to recover the latent components that maximize the independence and the non-stationarity across segments.

3 Data

To evaluate the Independent Innovation Analysis framework, we utilized two distinct types of datasets: a controlled synthetic dataset to validate the implementation and exploring the algorithm's sensitivity, and a real-world high-dimensional financial dataset to test the method's applicability to complex, non-stationary dynamics.

3.1 Simulated Data: NVAR Process

To reproduce the results of Morioka et al. and validate our implementation, we generated artificial data following a first-order Nonlinear Vector Autoregressive (NVAR) process. The innovations $\mathbf{s}_t$ were drawn from Gaussian distributions whose parameters (mean and variance) were modulated over time to satisfy the IIA assumption of non-stationarity. We used modulation functions based on Fourier bases (combinations of sine and cosine) to ensure smooth temporal evolution, making the data suitable for both GCL (continuous modulation) and TCL (segment-wise approximation). The observations $\mathbf{x}_t$ were generated by passing the concatenated vector $[\mathbf{x}_{t-1}, \mathbf{s}_t]$ through a random Multi-Layer Perceptron (MLP), creating a complex nonlinear mixing that simple linear methods cannot invert.

3.2 Real World Data: Financial Log-Returns

We collected daily prices of 10 major US stocks (AAPL, MSFT, AMZN, GOOGL, META, NVDA, TSLA, KO, PG, JPM) from January 2018 to December 2023 via Yahoo Finance. This 6-year window includes distinct market regimes: pre-COVID growth, the 2020 crash, and the 2022 inflationary downturn.

We computed log-returns $\mathbf{r}_t = \ln(\mathbf{p}_t) - \ln(\mathbf{p}_{t-1})$, standardized them, and applied PCA to reduce dimensionality to d components. The resulting heteroskedasticity (time-varying variance) corresponds exactly to the innovation modulation assumption of IIA.

For the auxiliary variable u_t: IIA-TCL uses segment indices with $L = 21$ days (one trading month), assuming piecewise stationarity; IIA-GCL uses the continuous time index, assuming smooth volatility evolution.

4 Results

4.1 Experiments with IIA-GCL (Continuous Modulation)

Simulation. We generated a 10-component NVAR time series ($T = 32{,}768$) with Gaussian innovations modulated by Fourier bases ($K = 64$), mixed through a 2-layer MLP.

Figure 1 shows the correlation matrix between true and estimated innovations after Hungarian algorithm alignment. IIA-GCL achieves a **mean correlation of** 0.969 (range: 0.944–0.983), confirming successful recovery of independent innovations up to permutation and sign. The low off-diagonal correlations ($|\rho_{\text{off}}| < 0.05$) validate the mutual independence of extracted components.

Application to financial data. We applied IIA-GCL to log-returns of 10 US stocks (2018–2023), reduced to $d = 3$ principal components, using the continuous time index as auxiliary variable.

Figure 4 shows the correlation matrix of the 3 extracted innovations. The off-diagonal correlations remain low (max $|\rho| = 0.16$), indicating approximate independence of the components.

The rolling variance plot (Fig. 5) reveals strong nonstationarity: all components exhibit a massive variance spike around $t \approx 550$, corresponding to the **COVID-19 market crash (March 2020)**. The coefficient of variation of segment variance ranges from 0.49 to 0.80, confirming pronounced heteroskedasticity.

Figure 6 shows correlations with the market factor (PC1). Innovation $\hat{s}_t^{(1)}$ exhibits weak directional correlation ($\rho = -0.18$) but strong correlation with absolute market movements ($\rho = 0.67$), suggesting it captures *volatility regimes* rather than directional trends. In contrast, $\hat{s}_t^{(0)}$ shows moderate directional correlation ($\rho = 0.13$) and lower volatility correlation ($\rho = 0.36$), indicating it partially captures *market-wide shocks*. Component $\hat{s}_t^{(2)}$ appears largely idiosyncratic with low correlations on both metrics.

The time series (Fig. 7) confirm these interpretations: during the COVID crisis, $\hat{s}_t^{(1)}$ shows symmetric positive spikes (volatility), while $\hat{s}_t^{(2)}$ exhibits a sharp negative shock (directional movement).

4.2 Experiments with IIA-TCL (Segment-wise Modulation)

Simulation and sensitivity to the number of segments n_{segment}. As previously, we generated a 10-component ($n = 10$) NVAR time series to evaluate the reconstruction performance of IIA-TCL. Figure 3 displays the Pearson correlation matrices between true and estimated innovations. The results confirm the theoretical requirement of having enough segments to guarantee identifiability (here $T \geq 21$).

- **With $n_{\text{segment}} = 256$:** The identifiability condition is met. We obtain a mean diagonal correlation of 0.989569. The matrix is clearly diagonal (Fig. 3a), indicating that each component is correctly isolated.
- **With $n_{\text{segment}} = 16$:** This parameter falls below the required theoretical threshold. Performance drops to a score of 0.390336. The diffuse structure of the matrix (Fig. 3b) illustrates the failure to disentangle the sources.

Thus, $n_{\text{segment}} = 256$ offers a satisfactory trade-off, ensuring both the modulation diversity necessary for identification and a sufficient number of samples per segment for training stability.

Application to financial data and model selection. We applied IIA-TCL to the financial dataset, setting the PCA dimension d equal to the number of sought components n. Comparing model orders reveals that $n = 3$ results in correlated components (Fig. 8a), indicating an undercomplete representation where distinct risk factors remain entangled. Conversely, increasing to $n = 5$ significantly improves diagonalization (Fig. 8b), suggesting sufficient degrees of freedom to isolate independent sources of non-stationarity.

Focusing on the $n = 5$ model, the segment-wise variance (Fig. 9) confirms the non-stationary hypothesis, revealing both systemic stress and component-specific fluctuations. This is exemplified by the time series during the **COVID-19 crash** ($t \approx 550$) in Fig. 10. Crucially, the method disentangles different "flavors" of the crisis: $\hat{s}_t^{(3)}$ captures a directional shock, whereas $\hat{s}_t^{(0)}$ and $\hat{s}_t^{(4)}$ exhibit symmetric volatility bursts, effectively separating trends from pure volatility regimes.

4.3 Conclusion

In summary, our experiments validate the ability of both frameworks to capture market non-stationarity. While **IIA-GCL** ($d = 3$) effectively isolates systemic risk factors via continuous modulation, **IIA-TCL** ($d = 5$) offers a more granular decomposition, distinguishing directional shocks from pure symmetric volatility bursts. A common limitation, however, remains the Gaussian assumption, which underestimates the heavy tails (leptokurtosis) of real returns, suggesting the benefit of extending the method to more robust distributions (e.g., Student-t).

References

[1] H. Morioka, H. Hälvä, and A. Hyvärinen. Independent Innovation Analysis for Nonlinear Vector Autoregressive Process. In *Proceedings of the 24th International Conference on Artificial Intelligence and Statistics (AISTATS)*, volume 130 of PMLR, 2021.

[2] C. A. Sims. Macroeconomics and reality. *Econometrica*, 48(1):1–48, 1980.

[3] A. Hyvärinen and P. Pajunen. Nonlinear independent component analysis: Existence and uniqueness results. *Neural Networks*, 12(3):429–439, 1999.

[4] A. Hyvärinen and H. Morioka. Unsupervised feature extraction by time-contrastive learning and nonlinear ICA. In *Advances in Neural Information Processing Systems (NIPS)*, pages 3765–3773, 2016.

[5] A. Hyvärinen, H. Sasaki, and R. Turner. Nonlinear ICA using auxiliary variables and generalized contrastive learning. In *Proceedings of the 22nd International Conference on Artificial Intelligence and Statistics (AISTATS)*, pages 859–868, 2019.

A Results on simulated data

A.1 IIA-GCL

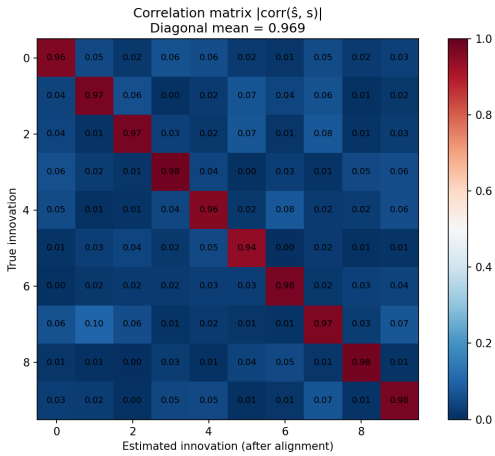


Figure 1: Correlation matrix $|\mathrm{corr}(\hat{s}, s)|$ between true and estimated innovations on simulated NVAR data (IIA-GCL). Diagonal mean = 0.969.

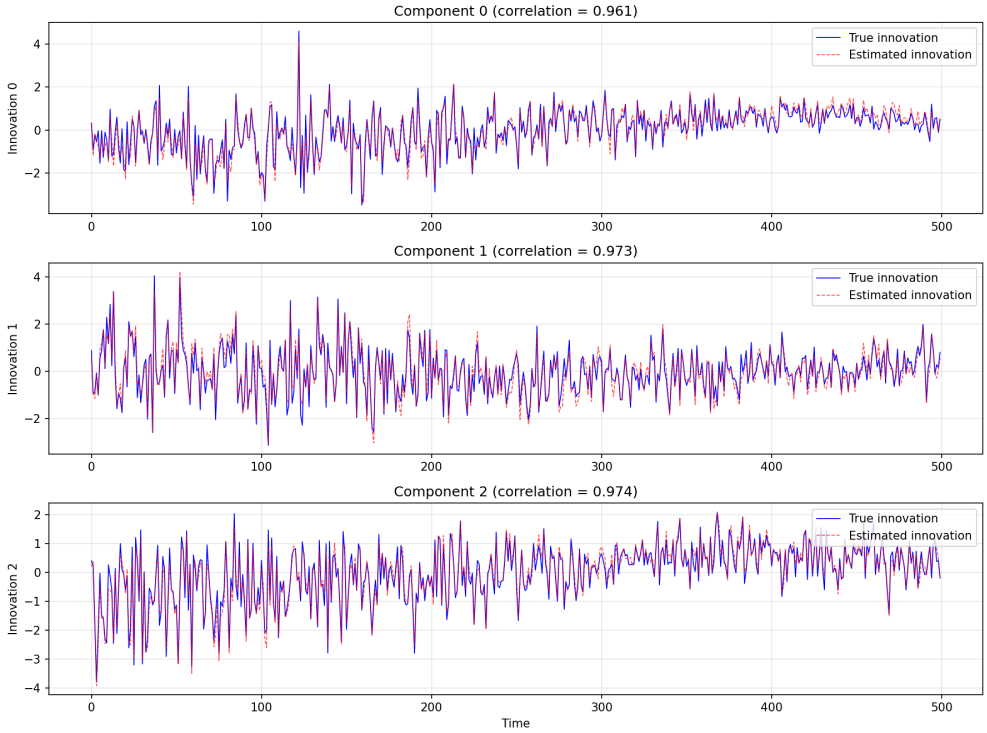
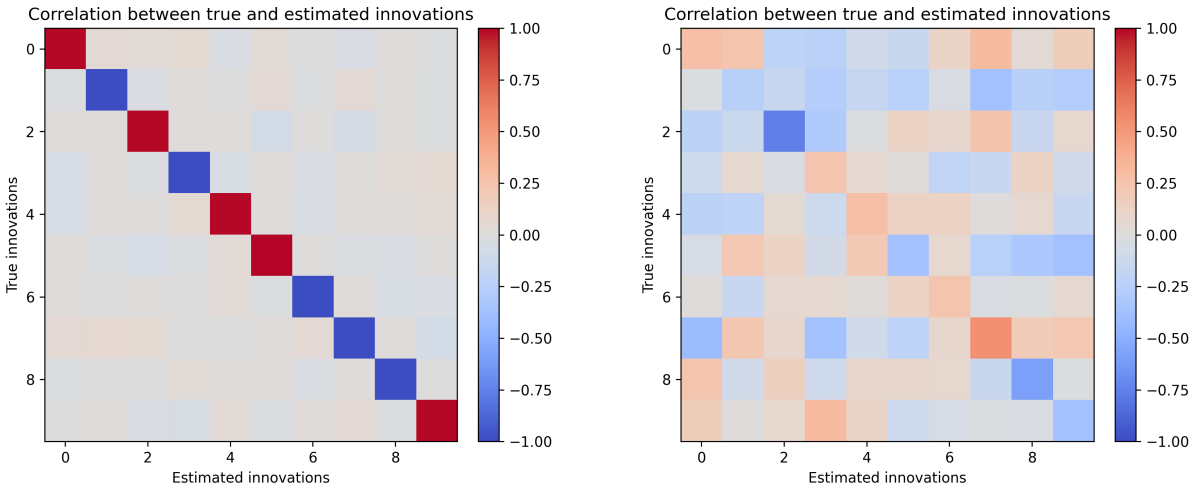


Figure 2: Temporal comparison of true (blue) and estimated (red) innovations for three components on simulated data.

A.2 IIA-TCL



(a) $n_{\text{segment}} = 256$ (b) $n_{\text{segment}} = 16$

Figure 3: Correlation matrix $|\text{corr}(\hat{s}, s)|$ between true and estimated innovations on simulated NVAR data (IIA-TCL), for different number of segments.

B Results on financial data

B.1 IIA-GCL

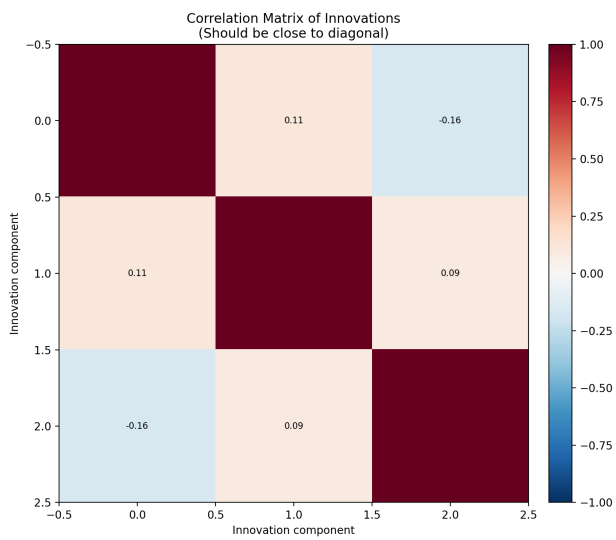


Figure 4: Correlation matrix of innovations extracted by IIA-GCL on financial data. Off-diagonal values remain low (max $|\rho| = 0.16$), indicating approximate independence.

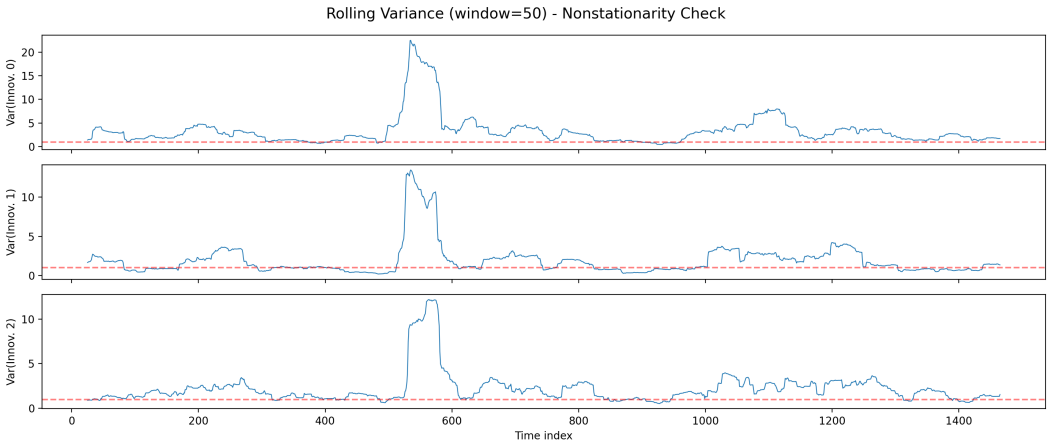


Figure 5: Rolling variance (window=50 days) of IIA-GCL innovations. The spike at $t \approx 550$ corresponds to the COVID-19 market crash (March 2020).

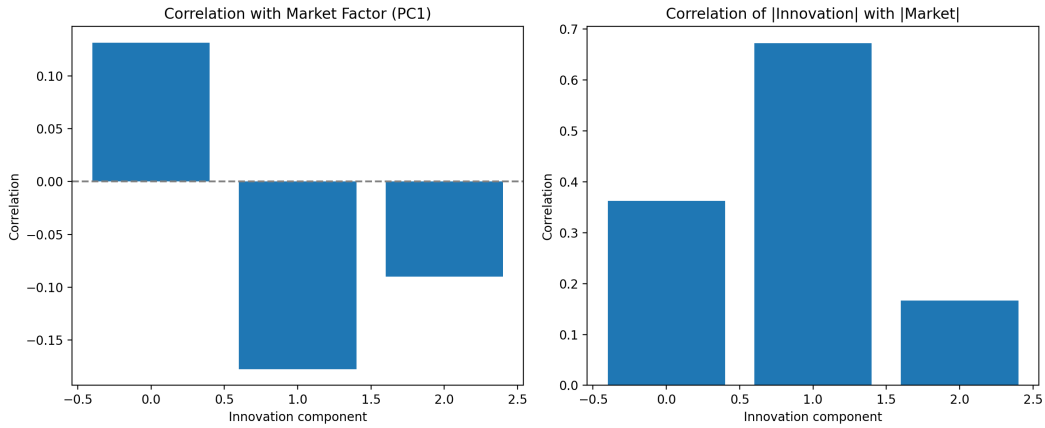


Figure 6: Correlation of IIA-GCL innovations with market factor (PC1). Left: directional correlation. Right: correlation of absolute values (volatility proxy).

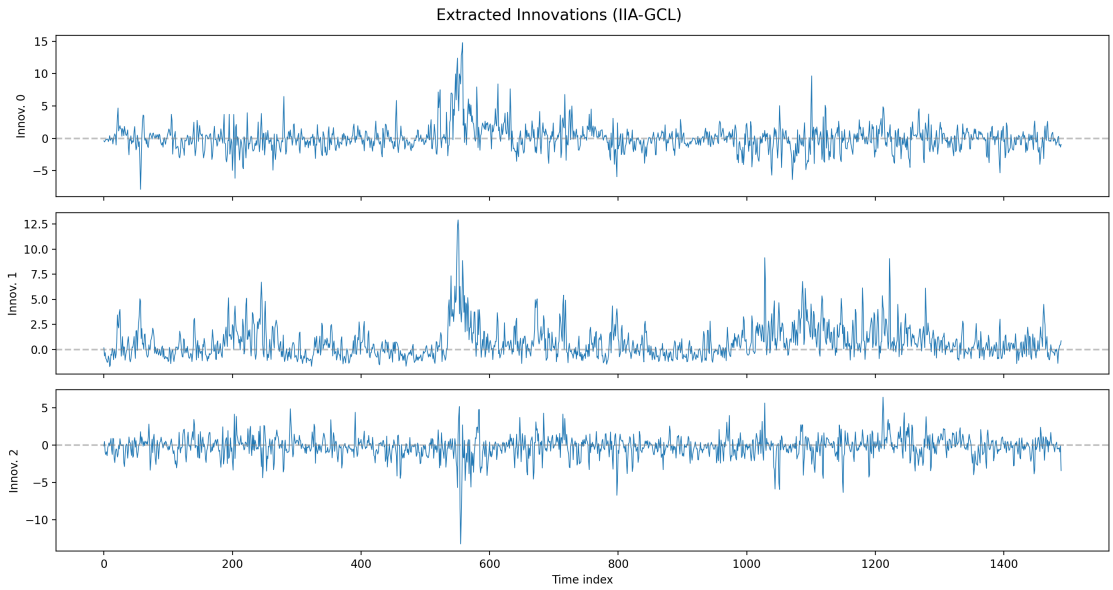
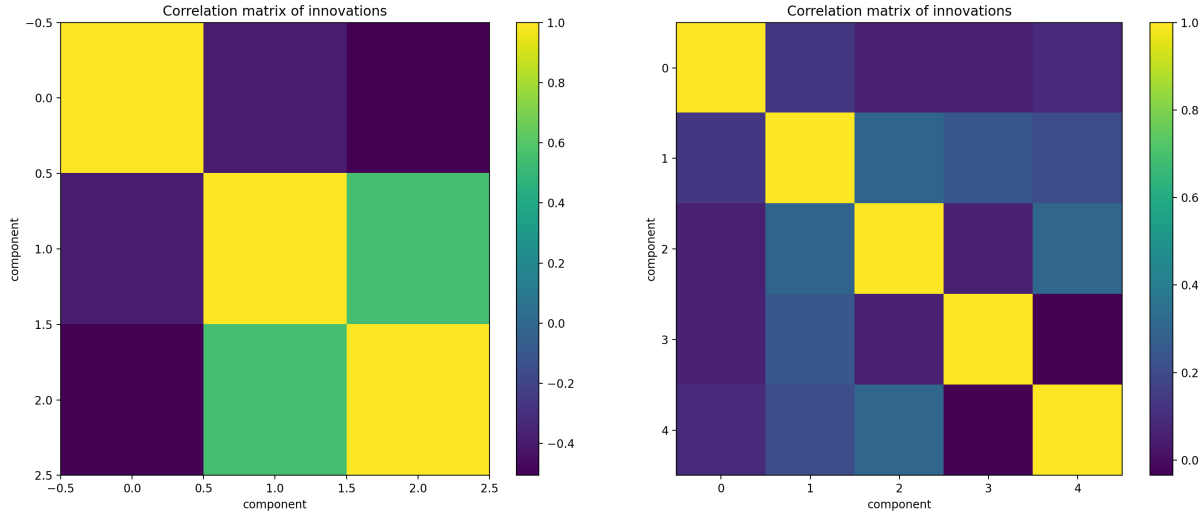


Figure 7: Time series of innovations extracted by IIA-GCL from financial data (2018–2023).

B.2 IIA-TCL



(a) $n_{\text{component}} = 3$

(b) $n_{\text{component}} = 5$

Figure 8: Correlation matrix of innovations extracted by IIA-TCL on financial data, for different number of component.

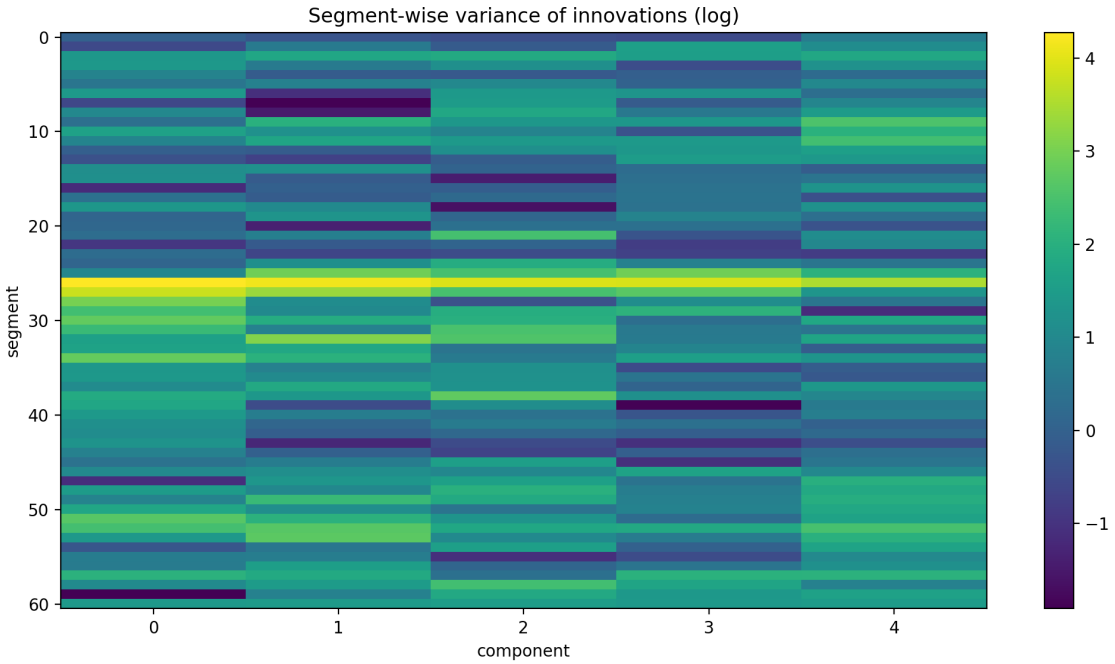
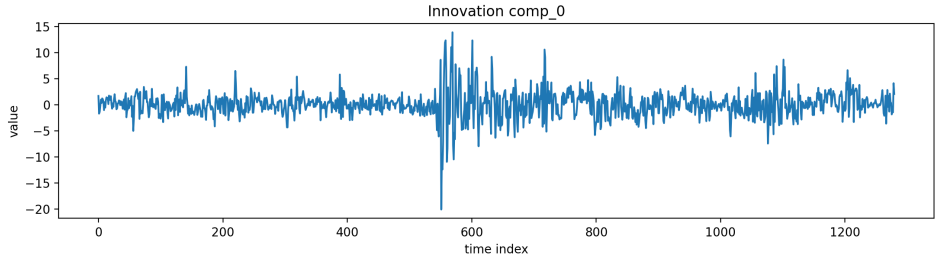
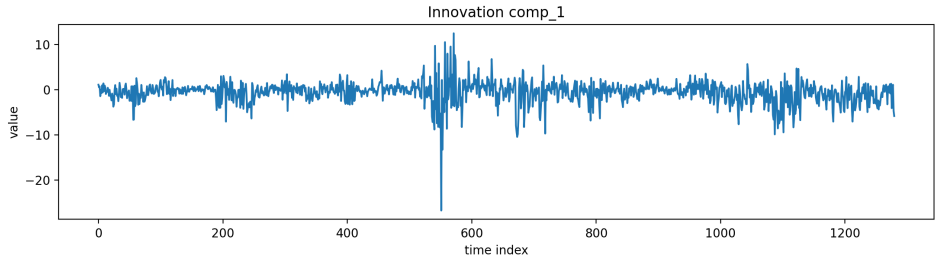


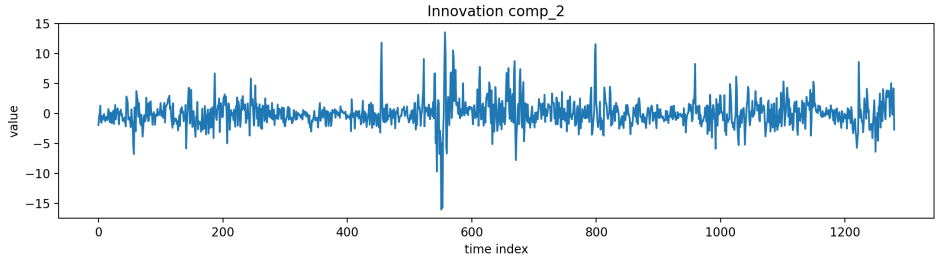
Figure 9: Segment-wise variance of innovations extracted by IIA-TCL on financial data (log scale). Each row corresponds to a segment (approximately 21 trading days) and each column to a component.



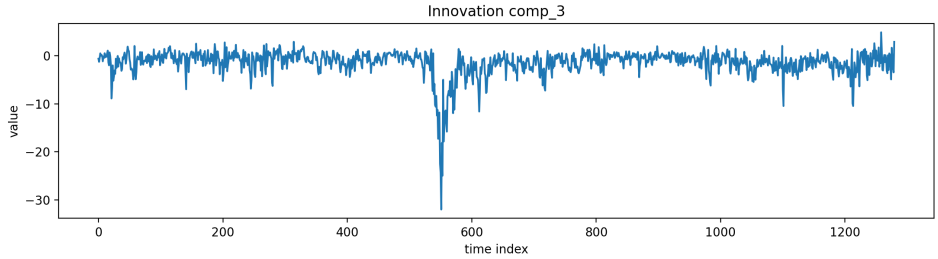
(a) Innovation $\hat{s}_t^{(0)}$



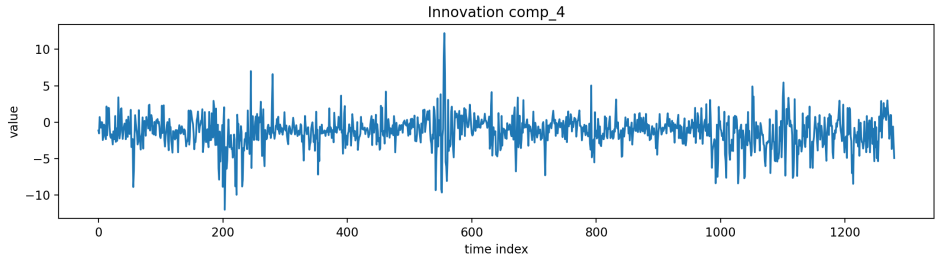
(b) Innovation $\hat{s}_t^{(1)}$



(c) Innovation $\hat{s}_t^{(2)}$



(d) Innovation $\hat{s}_t^{(3)}$



(e) Innovation $\hat{s}_t^{(4)}$

Figure 10: Estimated innovations.